Project Plan: Smart Barbell Load Cuffs

Assignments 5 and 6 · Task List, Milestones, Timeline and Effort Matrix
Team: Tyler Beck, Colin Hill
Sources: `Tasklist.md` (repository root), `Planning/Project_Plan.md`, `Planning/gantt.py` (draws `Gantt.png`). Task names use the component names of D0–D2.

Dates. Course weeks are counted from the D2 due date (Week 7 = Wednesday, October 14, 2026). Fall dates marked ⟨confirm⟩ and all spring 2027 dates must be checked against the Week 1 deliverable calendar and the spring academic calendar. The plan assumes:

- Thanksgiving: November 26–27.
- Exam week: December 7–11.
- Winter break: until January 10.
- CS 5002 starts January 11.
- Spring break: March 8–12.
- Expo: April 22.

1. Task List (Assignment #5)

Owners follow the D0 component table: Tyler owns the ring (C1, C2, C3) and calibration (G3); Colin owns the phone app, server and dashboard (C4, C5, C6) and the glue components G1, G2 and G4. Colin is the test lead and Tyler the documentation lead.

ID	Task (one statement, verb first)	Module	Owner
T1	Obtain the sprint parts and bench-test C1 Load sensing on a practice bar: bond the gauges, wire the full bridge to the HX711 and X1, and log a known load.	C1, X1	Tyler Beck
T2	Develop the C2 Ring firmware core: the I2 and I3 drivers, the I6 Bluetooth service, and the D2 Figure 2 state machine with Sleep and the 20-set flash buffer.	C2	Tyler Beck
T3	Develop algorithm A1 (side weight) in C2 together with the G3 calibration procedure, and test it to ±2 kg on the sprint build.	C2 (A1), G3	Tyler Beck
T4	Develop algorithm A2 (rep count and bar speed) in C2 and validate it against a reference measurement.	C2 (A2)	Tyler Beck
T5	Develop the Rev E amplifier board for C1 (KiCad layout, fabrication, assembly) and validate drift with excitation reversal.	C1	Tyler Beck
T6	Design the sensor ring housing for C1, C2 and C3 after measuring rec-center rack clearance, and print a fit-test pair.	C1–C3 housing	Tyler Beck
T7	Develop the C5 Account service data model (D1 Figure 1) as database migrations and deploy it on X4.	C5, X4	Colin Hill
T8	Develop the C5 session and set endpoints with session arbitration and idempotent uploads (D1 section 5.1, D2 section 2.5).	C5	Colin Hill
T9	Develop G1 Authentication with the X3 identity provider for C4 and C5.	G1, X3	Colin Hill
T10	Develop the C4 Lifter phone app tap-to-session flow and Bluetooth client (D2 Figure 4, interfaces I5 and I6).	C4	Colin Hill

ID	Task (one statement, verb first)	Module	Owner
T11	Develop C4 set assembly, the review screen and the upload queue, and G2 Notifications for bar handover (D2 Figure 3).	C4, G2	Colin Hill
T12	Develop the C6 Staff dashboard and G4 Logging and health checks.	C6, G4	Colin Hill
T13	Test the integrated system (C1–C6, G1–G4) against the Week 10 test plan and fix every Must-level defect.	All	Colin Hill
T14	Validate the system with lifters and a coach at the rec center, covering US-01 to US-04.	All	Tyler Beck
T15	Document the final report, poster and Expo demonstration for the complete system.	All	Tyler Beck

Fifteen tasks: eight for Tyler and seven for Colin. Each is one to two weeks of calendar time, with 10–27 expected hours.

2. Milestones List (Assignment #6)

Course deliverables (CD) are fixed by the course; **internal milestones** (M) are spaced every one to three weeks through the implementation semester, except across exam week and winter break.

ID	Milestone	What it demonstrates	Date	How it is verified	Verified by	Status
CD-1	Constraints essay	The eight constraint areas that shape the design	Week 3, Sep 16 ⟨confirm⟩	Submitted on Canvas	Instructor	**Complete**
CD-2	User stories and use cases	Stakeholders, US-01 to US-04, UC-01 and UC-02 with acceptance criteria	Week 4, Sep 23	Submitted; `User_Stories.md` in the repository	Instructor	**Complete**
CD-3	Design D0	Components, interfaces and architecture pattern	Week 5, Sep 30	Submitted; D0 files in `Design_Diagrams`	Instructor	**Complete**
CD-4	Design D1	Data model, core algorithms, API contract, technology choices	Week 6, Oct 7	Submitted; D1 files committed	Instructor	**Complete**
CD-5	Design D2 and peer review	Every component finished; 45 edge cases; traceability	Week 7, Oct 14	Consolidated Assignment 4 PDF; peer review notes	Instructor, reviewing teams	**Complete**
CD-6	Task list and plan (this document)	Owners, dates, effort and risks for the build	Week 8, Oct 21	Submitted; `Tasklist.md` and `Planning/` committed	Instructor, advisor	**Complete**
CD-7	Requirements Document and Technical Specification	Testable REQ and NFR IDs; traceability matrix	Week 9, Oct 28	Submitted after advisor review	Advisor, instructor	**Complete**
CD-8	Test Plan	Test cases tied to every Must requirement	Week 10, Nov 4	Submitted	Instructor	**Complete**
CD-9	Ethics Statement	Privacy, security, licensing, AI-use disclosure	Week 11, Nov 11	Submitted; signed by both members	Instructor	**Complete**
M1	Bench rig reads a known load	One side of the practice bar reports a 20 kg plate through the HX711 within ±2 kg, repeatably	Nov 24	Log of 10 load and unload cycles committed to `test-logs/`	Colin reviews Tyler's log	Planned
CD-10	Fall design presentation	The design and the plan, to the class	Week 13 ⟨confirm⟩	Presentation delivered	Instructor, advisor	Planned
CD-11	Final Design Report (Assignment #10)	The complete fall design package	Week 14, Dec 2 ⟨confirm⟩	Submitted	Instructor, advisor	Planned
M2	Server live; ring housing fits the real bar and rack	Sign-in works against the deployed C5; a printed ring clamps on the bar and clears the rack's J-hook	Dec 4	Screenshot of a signed-in API call; photo and measurement of the ring on a rec-center rack	Tyler checks Colin's deploy; Colin checks the fit	Planned
M3	Ring and server each work alone	The ring sends WEIGHT, REP and STATUS events to a generic BLE app; the API passes its scripted request collection	Jan 29	Screen recording; automated API test run	Each member checks the other's	Planned

ID	Milestone	What it demonstrates	Date	How it is verified	Verified by	Status
M4	Tap to session, end to end	A tap opens the app, signs in, opens a session and binds both rings	Feb 17	20 timed taps, at least 19 within 5 s (AC-01.2)	Advisor demo ⟨confirm⟩	Planned
M5	Working prototype	A set is logged automatically from tap to account with the Rev E amplifier fitted	Mar 6	60 kg reference load, 5 reps, saved within ±1.0 kg (AC-01.1 rehearsal)	Advisor	Planned
M6	Speed validated; dashboard live	Rep count and bar speed match a reference; the coach and equipment pages show real data	Mar 26	Comparison table against the reference; dashboard walkthrough	Colin checks speeds; Tyler checks the dashboard	Planned
M7	All Must tests pass	The integrated system meets every Must requirement	Apr 9	Week 10 test plan executed, results logged	Advisor	Planned
M8	Pilot complete	Real lifters and a coach used the bar	Apr 16	Pilot log and questionnaire results	Advisor	Planned
M9	Expo demonstration	The finished system, presented	Apr 22 ⟨confirm⟩	Live demo, poster, final report	Judges, advisor	Planned

3. Table 1: Timeline (Assignment #6)

Every finish date already includes the task's own buffer: about 15–20% of its length, shown as the last days of the window. **PB** rows are protected blocks before major milestones. No task work is planned in them; they absorb slips. **Bold** IDs are on the critical path:

T1 → T2 → T3 → T5 → T4 → T13 → T15

The hardware chain is critical because T5 waits for PCB fabrication. Colin's chain (T7 → T9 → T8 → T10 → T11) runs in parallel and ends on the same dates. A slip in either chain uses up PB-2 before it moves M5.

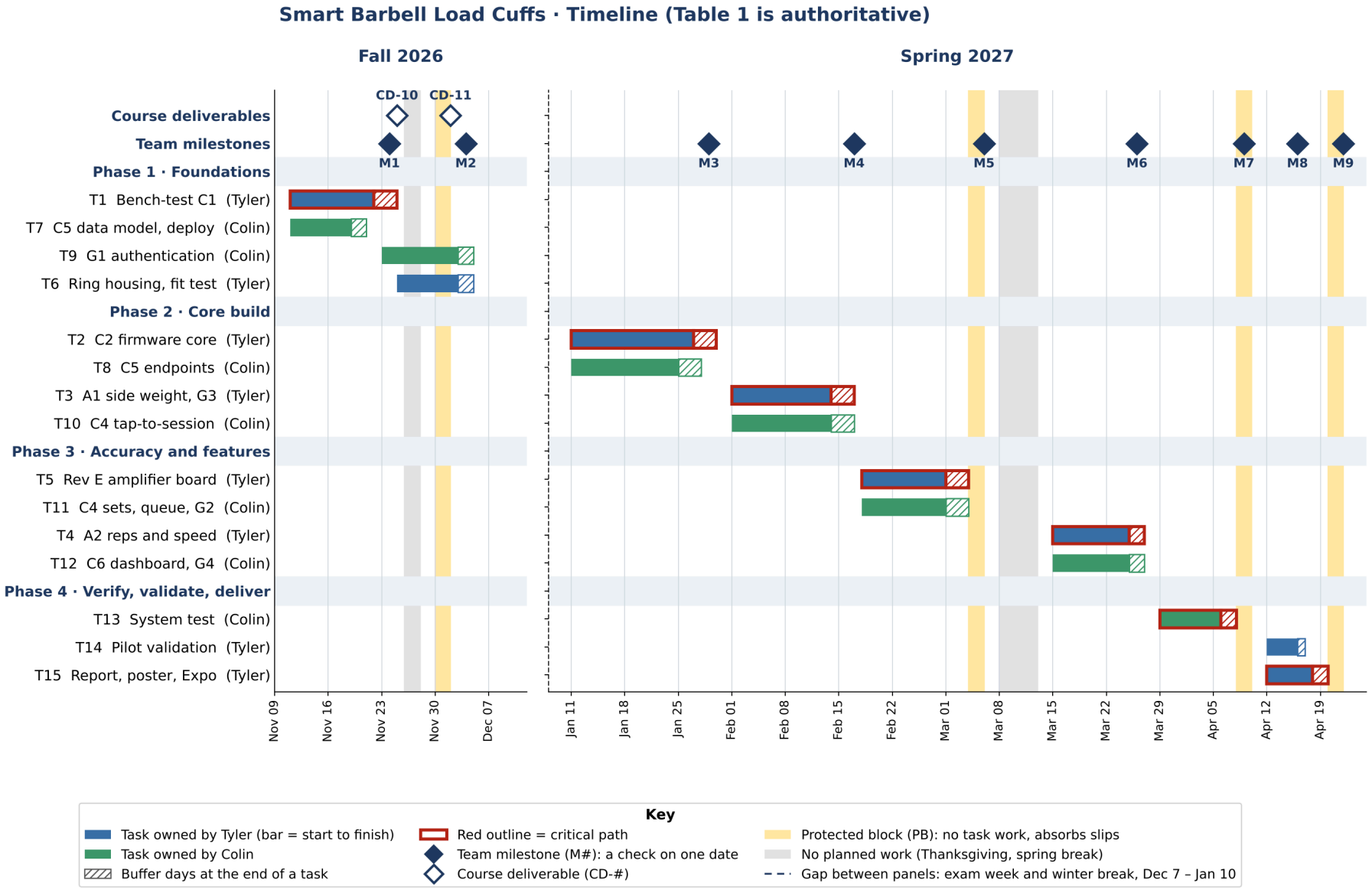
ID	Task	Owner	Start	Finish (with buffer)	Buffer inside	Depends on	Schedule risk
T1	Bench-test C1 Load sensing	Tyler	Wed Nov 11, 2026	Tue Nov 24	3 days (Nov 22–24)	Parts delivered	RK-1, RK-3 (parts late)
T7	C5 data model and deployment	Colin	Wed Nov 11	Fri Nov 20	2 days (Nov 19–20)	—	RK-7
T9	G1 Authentication	Colin	Mon Nov 23	Fri Dec 4	2 days (Dec 3–4)	T7	—
T6	Ring housing and fit test	Tyler	Wed Nov 25	Fri Dec 4	2 days (Dec 3–4)	T1 (gauge positions); rack measured	**RK-4**
PB-1	Protected block before CD-11	Both	Mon Nov 30	Tue Dec 1	—	—	—
—	Exam week and winter break: no planned work	—	Mon Dec 7	Sun Jan 10, 2027	—	—	RK-6
T2	C2 firmware core	Tyler	Mon Jan 11, 2027	Fri Jan 29	3 days (Jan 27–29)	T1	RK-5
T8	C5 session and set endpoints	Colin	Mon Jan 11	Wed Jan 27	3 days (Jan 25–27)	T7, T9	—
T3	A1 side weight and G3 calibration	Tyler	Mon Feb 1	Tue Feb 16	3 days (Feb 14–16)	T2	**RK-1**
T10	C4 tap-to-session and Bluetooth client	Colin	Mon Feb 1	Tue Feb 16	3 days (Feb 14–16)	T8, T2	**RK-2**
T5	Rev E amplifier board	Tyler	Thu Feb 18	Wed Mar 3	3 days (Mar 1–3)	T3; boards ordered by Feb 1	**RK-3**
T11	C4 sets, upload queue, G2	Colin	Thu Feb 18	Wed Mar 3	3 days (Mar 1–3)	T10, T3	RK-2
PB-2	Protected block before M5	Both	Thu Mar 4	Fri Mar 5	—	—	—
—	Spring break: no planned work	—	Mon Mar 8	Fri Mar 12	—	—	—

ID	Task	Owner	Start	Finish (with buffer)	Buffer inside	Depends on	Schedule risk
T4	A2 reps and speed	Tyler	Mon Mar 15	Fri Mar 26	2 days (Mar 25–26)	T2, T5	RK-1
T12	C6 dashboard and G4	Colin	Mon Mar 15	Fri Mar 26	2 days (Mar 25–26)	T8	—
T13	System test	Colin	Mon Mar 29	Wed Apr 7	2 days (Apr 6–7)	T4, T11, T12	RK-5
PB-3	Protected block before M7	Both	Thu Apr 8	Fri Apr 9	—	—	—
T14	Pilot validation	Tyler	Mon Apr 12	Fri Apr 16	1 day (Apr 16)	T13 (M7)	**RK-8**
T15	Report, poster, Expo	Tyler	Mon Apr 12	Mon Apr 19	2 days (Apr 18–19)	T13; T14 results	—
PB-4	Protected block before M9	Both	Tue Apr 20	Wed Apr 21	—	—	—

Milestone completion dates

Milestone	Planned completion	Protected block before it
M1 Bench rig reads a known load	Tue Nov 24, 2026	T1's 3-day buffer
CD-11 Final Design Report	Wed Dec 2	PB-1
M2 Server live; housing fits	Fri Dec 4	Task buffers
M3 Ring and server each work alone	Fri Jan 29, 2027	Task buffers
M4 Tap to session, end to end	Wed Feb 17	Task buffers
M5 Working prototype	Sat Mar 6	PB-2
M6 Speed validated; dashboard live	Fri Mar 26	Task buffers
M7 All Must tests pass	Fri Apr 9	PB-3
M8 Pilot complete	Fri Apr 16	T14's buffer
M9 Expo demonstration	Thu Apr 22	PB-4

Gantt figure (supporting; Table 1 is authoritative). Bars are tasks: each runs from its start to its finish date, and the hatched end is its buffer. Diamonds are milestones, single dates when something is checked. The key under the chart explains every color and mark.


Smart Barbell Load Cuffs · Timeline (Table 1 is authoritative)
Fall 2026
Spring 2027
CD-10
CD-11
Course deliverables
Team milestones
M1
M2
M3
M4
M5
M6
M7
M8
M9
Phase 1 · Foundations
T1 Bench-test C1 (Tyler)
T7 C5 data model, deploy (Colin)
T9 G1 authentication (Colin)
T6 Ring housing, fit test (Tyler)
Phase 2 · Core build
T2 C2 firmware core (Tyler)
T8 C5 endpoints (Colin)
T3 A1 side weight, G3 (Tyler)
T10 C4 tap-to-session (Colin)
Phase 3 · Accuracy and features
T5 Rev E amplifier board (Tyler)
T11 C4 sets, queue, G2 (Colin)
T4 A2 reps and speed (Tyler)
T12 C6 dashboard, G4 (Colin)
Phase 4 · Verify, validate, deliver
T13 System test (Colin)
T14 Pilot validation (Tyler)
T15 Report, poster, Expo (Tyler)
Nov 09
Nov 16
Nov 23
Nov 30
Dec 07
Jan 11
Jan 18
Jan 25
Feb 01
Feb 08
Feb 15
Feb 22
Mar 01
Mar 08
Mar 15
Mar 22
Mar 29
Apr 05
Apr 12
Apr 19
Key
Task owned by Tyler (bar = start to finish)
Task owned by Colin
Buffer days at the end of a task
Red outline = critical path
Team milestone (M#): a check on one date
Course deliverable (CD-#)
Protected block (PB): no task work, absorbs slips
No planned work (Thanksgiving, spring break)
Gap between panels: exam week and winter break, Dec 7 – Jan 10

4. Table 2: Effort Matrix (Assignment #6)

Each owner gave three estimates (optimistic O, most likely M, pessimistic P). The expected hours are E = (O + 4M + P) / 6, rounded. The ★ marks the member of primary responsibility, who carries the largest share of every row. The support shares are reviews, pair debugging and hand-offs at interfaces.

ID	Task	O	M	P	E (h)	Tyler (h)	Colin (h)
T1	Bench-test C1	8	12	20	13	★ 10	3
T2	C2 firmware core	16	24	40	25	★ 22	3
T3	A1 side weight, G3	12	20	36	21	★ 18	3
T4	A2 reps and speed	12	20	36	21	★ 18	3
T5	Rev E amplifier board	10	16	30	17	★ 15	2
T6	Ring housing, fit test	8	14	24	15	★ 10	5
T7	C5 data model, deploy	6	10	16	10	1	★ 9
T8	C5 endpoints	14	22	36	23	2	★ 21
T9	G1 Authentication	6	10	20	11	1	★ 10
T10	C4 tap-to-session	16	26	44	27	5	★ 22
T11	C4 sets, queue, G2	14	22	36	23	3	★ 20
T12	C6 dashboard, G4	10	16	26	17	2	★ 15
T13	System test	12	18	30	19	8	★ 11
T14	Pilot validation	10	16	28	17	★ 10	7
T15	Report, poster, Expo	16	24	36	25	★ 14	11
Total					**284**	**139 (49%)**	**145 (51%)**

Balance. The columns are within 6 hours of each other (49% and 51%), about 7 hours per week each across the remaining 20 working weeks. The risk is not the totals but the critical path: most of Tyler's hours sit on it, so a slip in T3 or T5 delays the whole project.

What we will do about it:

- Colin takes the supporting share of T4 and T13 early if Tyler's tasks slip past a buffer.
- Each Friday we compare actual hours with this table. If either member is more than 15% over plan for two weeks in a row, we move the support share of T15 (the poster and the demo script) to the other member and log the change in this document.

5. Work Breakdown Structure

- **1 Foundations** (fall, Weeks 11–14)
  - T1 Bench-test C1 Load sensing (Tyler)
  - T6 Ring housing and fit test (Tyler)
  - T7 C5 data model and deployment (Colin)
  - T9 G1 Authentication (Colin)
- **2 Core build** (spring, Weeks 1–5)
  - T2 C2 firmware core (Tyler)
  - T3 A1 side weight and G3 calibration (Tyler)
  - T8 C5 session and set endpoints (Colin)
  - T10 C4 tap-to-session and Bluetooth client (Colin)
- **3 Accuracy and features** (spring, Weeks 6–11)
  - T5 Rev E amplifier board (Tyler)
  - T4 A2 reps and speed (Tyler)
  - T11 C4 set assembly, upload queue and G2 (Colin)
  - T12 C6 dashboard and G4 (Colin)
- **4 Verification and validation** (spring, Weeks 12–14)
  - T13 System test (Colin)
  - T14 Pilot validation (Tyler)
- **5 Delivery** (spring, Weeks 14–15)
  - T15 Report, poster and Expo (Tyler)

6. Risk Register

Likelihood and impact are Low, Medium or High. Risks marked in Table 1 are the high-impact schedule risks.

ID	Category	Risk	Likelihood	Impact	Strategy	Mitigation	Owner	Trigger
RK-1	Technical	Gauges bonded with the sprint glue drift or debond, so side weight misses ±1.0 kg (NFR-01)	Medium	High	Reduce	Practice bonds on scrap first; excitation reversal on the Rev E board (T5); AE-10 epoxy for the final bar; 60-minute drift test in T3	Tyler	Drift above 1 kg in the T3 drift test, or a gauge reading frozen under load
RK-2	Team (skills)	Android and Kotlin are new, so the phone app (T10, T11) is slower to build than planned	Medium	High	Reduce	Android Bluetooth tutorial over winter break; Nordic's BLE library instead of raw APIs; keep the Web Bluetooth test page as a demo fallback	Colin	Tap-to-session not working by Feb 10 (inside T10's buffer)
RK-3	External	PCB fabrication or shipping takes more than two weeks, delaying T5 on the critical path	Medium	Medium	Reduce	Order boards by Feb 1, before T5 starts; build the first amplifier on MSOP adapter boards so testing does not wait	Tyler	Boards not shipped by Feb 10
RK-4	External	Rec-center racks leave less than about 28 mm beside the J-hook, so the ring does not fit (NFR-08)	Medium	High	Avoid	Measure every rack before printing (T6, by M2); fall back to a slimmer ring or named racks only	Tyler	Any measured rack under 28 mm per side
RK-5	Team	One of two members is unavailable for one to two weeks (illness, interviews, co-op ⟨confirm co-op schedules⟩)	Medium	High	Reduce	Weekly code review of each other's work so either can continue; task buffers and protected blocks; the support shares in Table 2	Both	A member misses two weekly check-ins
RK-6	Team (schedule)	Exam week and winter break stop progress for five weeks	High	Medium	Accept	No tasks are planned from Dec 7 to Jan 10; optional break work is Android ramp-up only (RK-2)	Colin	— (planned)
RK-7	External	Hosting price or free-tier changes, or an outage	Low	Low	Transfer	Managed service; standard PostgreSQL migrations move to any host	Colin	Pricing notice or more than 1 h of outage
RK-8	External	The rec center does not allow a pilot, or student testing needs approval we have not asked for	Medium	Medium	Reduce	Ask the rec center and advisor in Week 12; fall back to a lab pilot with volunteers	Tyler	No written approval by Jan 29
RK-9	Technical	A LiPo cell is damaged when a bar is dropped	Low	High	Reduce	Protected cells in a foam cradle; impact tests in T13; never charge a damaged ring	Tyler	Any swollen or dented cell